

Lab-4

1. Data Extraction – CSV

You are given a file sales.csv containing the columns:
order_id, order_date, customer_id, product, quantity, price

Write a Python program to:

- Load the CSV file using pandas
- Display the first 5 rows
- Print total number of records

2. Data Extraction – JSON

A file customers.json contains customer details.

Write a Python script to:

- Read the JSON file
- Convert it into a pandas DataFrame
- Display only customer_id, name, and city

3. Data Cleaning (Transformation)

Using the sales.csv DataFrame:

- Remove rows where quantity or price is missing
- Convert order_date into YYYY-MM-DD format
- Add a new column total_amount = quantity * price

4. Data Filtering

From the transformed sales data:

- Filter all orders where total_amount > 5000
- Sort the result in descending order of total_amount

5. Aggregation for Analytics

Write a Python program to:

- Calculate total sales per product
- Calculate total sales per customer_id

(Store results in separate DataFrames)

6. ETL Pipeline – Modular Design

Design an ETL pipeline using **functions**:

- extract_data() – reads CSV file
- transform_data() – cleans and adds derived columns
- load_data() – saves processed data to a new CSV

Call these functions in sequence.

7. Loading Data into SQLite (Load Phase)

Write a Python program to:

- Create a SQLite database sales.db
- Create a table processed_sales
- Insert transformed sales data into the table

8. Data Validation

After loading data into SQLite:

- Fetch total number of records from the table
- Verify that it matches the transformed DataFrame row count

9. Logging ETL Steps

Enhance your ETL pipeline to:

- Print messages for each step: Extract, Transform, Load
- Display start time and end time of the pipeline

10. Applied Data Engineering Task

Write a Python script that:

- Automatically reads all CSV files from a folder raw_data/

- Merges them into a single DataFrame
- Applies cleaning and transformation
- Saves final output as final_dataset.csv